Koç University

College of Engineering

MECH 301 Fluid Mechanics

CFD Project I

Flow Through an Expansion in a Circular Pipe

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83155

1. Introduction

Fluid flow through pipes with varying cross-sections is a problem in fluid mechanics with applications in engineering systems. Specifically, flow through a sudden expansion involves complex flow phenomena such as boundary layer separation, the formation of recirculation zones and flow re-attachment. These phenomena result in irreversible energy dissipation, referred to as minor head losses, which are critical considerations in the design of piping networks, chemical reactors, and HVAC systems.

The primary objective of this project is to analyze the internal flow of a viscous fluid through a sudden expansion in a circular pipe using Computational Fluid Dynamics (CFD). Understanding the physics of flow separation and predicting the re-attachment length are needed for optimizing system efficiency and minimizing pressure losses.

Several experimental and numerical studies have been conducted to characterize sudden expansion flows. For instance, Macagno and Hung (1967) investigated the behavior of captive annular eddies in laminar flow, establishing a relationship between the Reynolds number and the re-attachment length [1]. More recently, Hammad et al. (1999) performed Particle Image Velocimetry measurements to visualize the velocity field and validate numerical predictions for laminar axisymmetric sudden expansion [2]. These studies serve as a benchmark for validating modern numerical simulations.

In this work, the commercial CFD package ANSYS Fluent is used to simulate the flow field based on the Finite Volume Method. The project is divided into three main tasks. First, a laminar flow regime $Re = 55.4$ is simulated to perform a grid convergence study and validate the numerical results against the analytical Poiseuille solution and experimental data from Hammad et al. (1999) . A parametric study is also conducted to observe the variation of the re-attachment length with the Reynolds number. Second, fully turbulent flow $Re = 10^5$ is analyzed

using two different Reynolds-Averaged Navier-Stokes turbulence models, namely the Standard $k-\epsilon$ and Standard $k-\omega$ models, to compare their accuracy in predicting flow separation. Finally, the pressure loss coefficients are evaluated for various expansion ratios and geometries to assess the impact of geometric design on energy efficiency.

The remainder of this report is organized as follows: Section 2 describes the physical problem, computational domain, and governing equations. Section 3 presents the results and discussions for laminar flow verification, turbulence modeling comparison, and pressure loss analysis. Finally, Section 4 summarizes the key findings and conclusions drawn from the study.

2. Problem Description

In this project, the flow of a viscous fluid through a sudden expansion in a circular pipe is analyzed. The primary objective is to investigate the flow separation and re-attachment phenomena caused by the sudden change in geometry. The flow is assumed to be steady, laminar, incompressible, and axisymmetric.

2.1 Geometry

The computational domain represents a 2D axisymmetric slice of the pipe. The geometry consists of a narrow inlet pipe followed by a sudden expansion into a wider pipe. Computational geometry was created using ANSYS DesignModeler. Since the flow is axisymmetric, a 2D planar surface was generated to represent a longitudinal slice of the pipe.

Sketching: First, the profile of the sudden expansion was sketched on the XY plane using the line tool. The dimensions were constrained according to the project specifications $(R_1 = 5mm,\ R_2 = 10mm,\ L_1 = 100mm,\ L_2 = 250mm)$

Surface Generation: The sketch was converted into a surface body using the Surfaces from Sketches feature.

Domain Type: The geometry was defined as a “Fluid" body to ensure the solver treats it as a fluid domain.

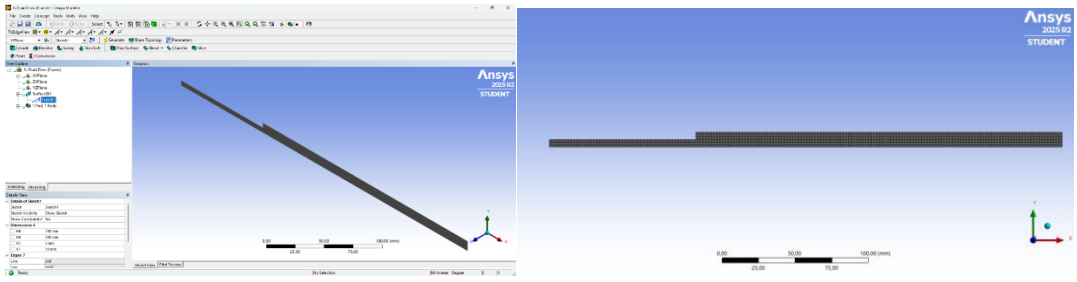


Figure 1: Sketch of the body in DesignModeler and 1mm coarse mesh.

By utilizing this body shown in Figure 1 on DesignModeler, It could be imported to any ansys file and dimensions could be changed or appended a parameter any time to calculate the pressure and velocity profiles. Also note that the axis of the part is in X direction. This sketch used in both part 1 and part 2 however it changed in part 3 using the parameter function of Ansys.

2.2 Material Properties and Flow Conditions

For the first task (Laminar Flow), the fluid is defined as air with modified properties to match the specific Reynolds number of $Re = 55.4$. The physical properties and flow parameters are defined as follows:

- Fluid: Air (Modified)
- Density (p): $100 kg/m^3$
- Dynamic Viscosity ($kg/(m*s)$: 0.01 ($kg/(m*s)$

2.3 Boundary Conditions

The computational domain is bounded by the inlet, outlet, walls, and the axis of symmetry. The following boundary conditions were applied in ANSYS Fluent:

- Inlet: A Velocity Inlet boundary condition is used with a constant axial velocity of V = 0.554 m/s.
- Outlet: A Pressure Outlet boundary condition is applied with a gauge pressure of 0 Pa.

- Wall: A No-Slip (Stationary Wall) condition is applied to the pipe walls.
- Axis: An Axisymmetric boundary condition is applied along the centerline (y=0).

2.4 Governing Equations

Since the flow is assumed to be steady, laminar, incompressible, and axisymmetric, the governing equations are the Continuity and Navier-Stokes equations [3] in cylindrical coordinates.

Continuity Equation:

$$\frac{\partial u}{\partial x} + \frac{1}{r}\frac{\partial (rv)}{\partial r} = 0$$

$Axial$ Momentum Equation:

$$\rho\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial r}\right) = -\frac{\partial p}{\partial x} + \mu\left[\frac{\partial^2 u}{\partial x^2} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)\right]$$

Radial Momentum Equation:

$$\rho\left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial r}\right) = -\frac{\partial p}{\partial r} + \mu\left[\frac{\partial^2 v}{\partial x^2} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v}{\partial r}\right) - \frac{v}{r^2}\right]$$

2.5 Task 2: Turbulent Flow Settings

In the second task, the Reynolds number was increased to Re = 10^5 to simulate fully turbulent flow. To achieve this without compressibility effects, the viscosity was significantly reduced while keeping the density constant. Two different Reynolds-Averaged Navier-Stokes (RANS) turbulence models were compared.

- **Target Reynolds Number:** $Re = 100{,}000$
- **Fluid Density (ρ):** $1000\ kg/m^3$
- **Dynamic Viscosity** : 0.0001
- **Inlet Velocity** ($\boldsymbol{V_inlet}$): $10m/s$
- **Turbulence Models Used:**

1. **$k-e$ Model:** Used with Standard Wall Functions.
2. **Standard $k-\omega$ Model:** Used for better resolution of the near-wall region.

Turbulence Intensity: Defined as 5% $at\ the\ inlet\ with\ a\ hydraulic\ diameter\ of$ 0.01 m.

2.6 Task 3: Geometric Variations (Pressure Loss)

In the final task, the effects of the expansion ratio ($\frac{d}{D}$) and expansion angle on the pressure loss coefficient (K) were investigated. The flow parameters from Task 1 (Laminar) were used to simulate these cases.

The following geometric variations were created and analyzed:

1. **Sudden Expansion Cases:**
   - $d/D = 0.6 (R_1 = 5mm, R_2 \approx 8.33mm)$
   - $d/D = 0.4 (R_1 = 5mm, R_2 = 12.5mm)$
   - $d/D = 0.2 (R_1 = 5mm, R_2 = 25mm)$

Table of Design Points

	A	B
1	Name	P1 - outlet.radius
2	Units	m
3	DP 0 (Current)	0,01
4	DP 1	0,0083
5	DP 2	0,0125
6	DP 3	0,025

Figure 2: Parameters tab of Ansys Workbench for Task 3 A column is cases and B column is outlet radius (R_2) .

2. **Gradual Expansion Case:**
   - Expansion Ratio: d/D = 0.4
   - Expansion Angle: $2\theta = 20°$ (Conical diffuser).

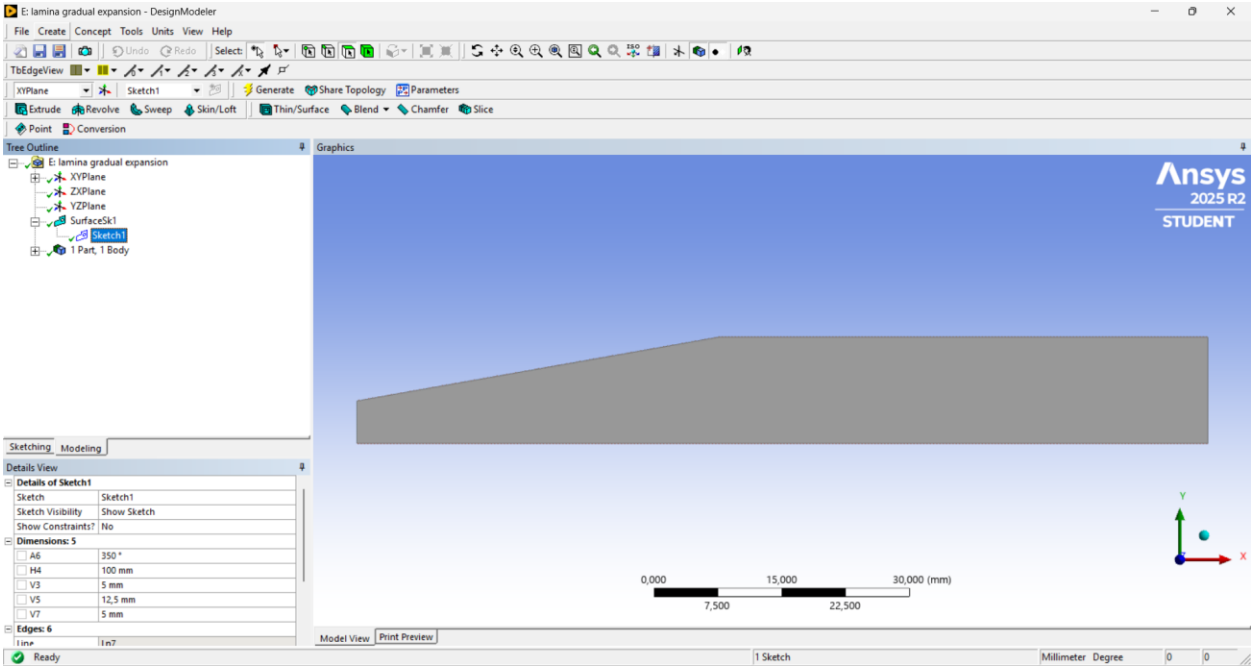


Figure 3: Body that is used in Task 3 gradual expansion case.

3. Results and Discussion

3.1 Task 1: Laminar Flow

In this section, the numerical results for the laminar flow case with a Reynolds number of Re = 55.4 are presented. The analysis includes a grid independence study, qualitative visualization of the flow field, and quantitative validation against analytical and experimental data.

3.1.1 Computational Grid and Convergence Study

To ensure that the numerical results are independent of the mesh resolution, a grid convergence study was performed. Three different structured grids (Coarse 1mm, Medium 0.5mm, and Fine 0.3mm) were generated. The element size was systematically reduced to refine the mesh.

The axial velocity profiles were monitored at five critical axial locations:

x = $10R_1, 20R_1, 21R_1, 25R_1, and\ 60R_1$.

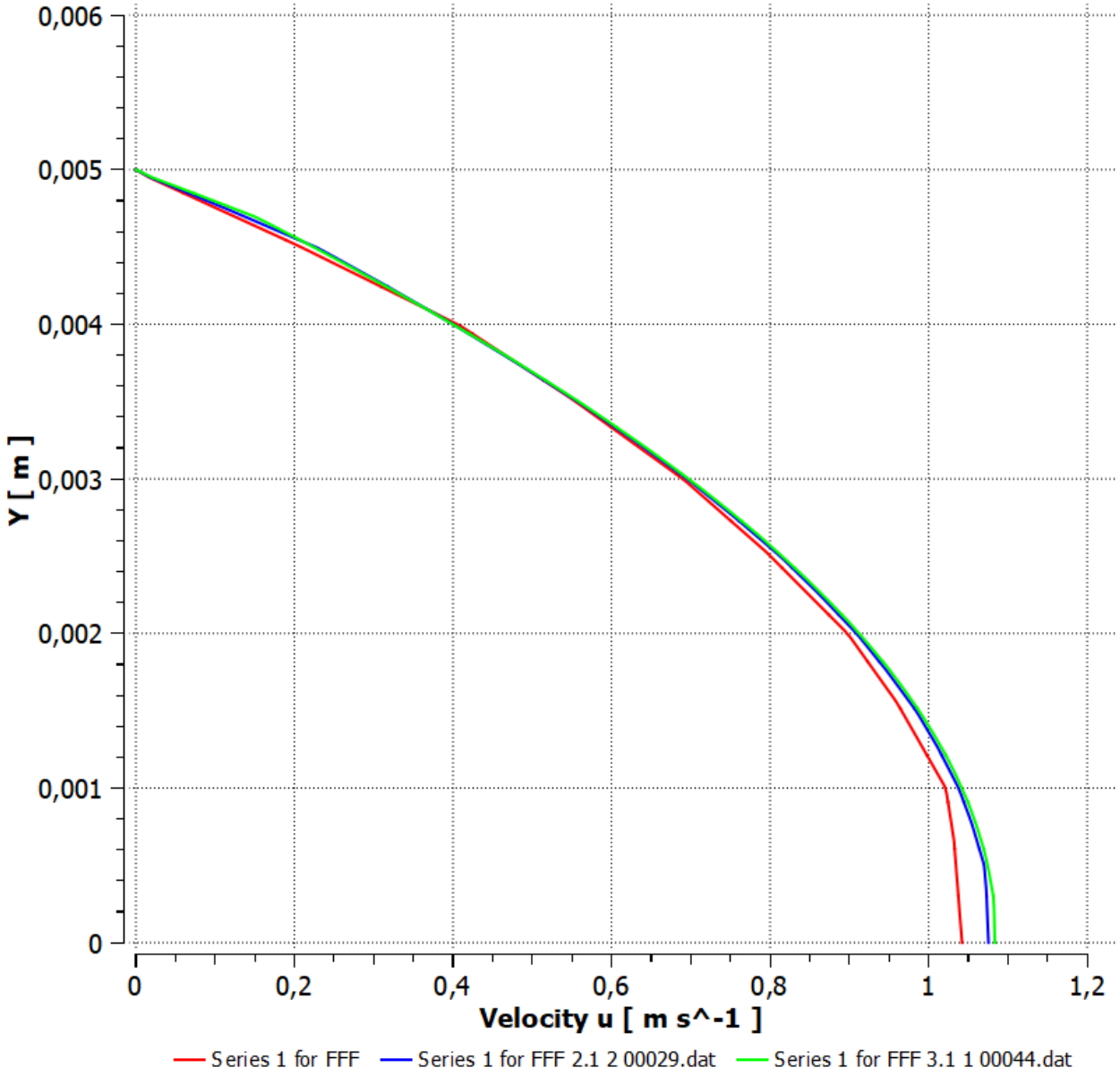


Figure 4. Comparison of axial velocity profiles for Coarse (Red), Medium (Blue), and Fine(Green) grids for x = -0.01.

As shown in Figure 4, the velocity profiles for the Medium and Fine grids are nearly identical, indicating that grid convergence has been achieved. So we can continue task without worrying about convergence of the task. The Fine mesh was selected for the subsequent analyses to ensure high accuracy. And also note that this is the first velocity field and 4 more will be given.

3.1.2 Qualitative Flow Analysis

The flow physics were analyzed using pressure contours, velocity vectors, and streamlines.

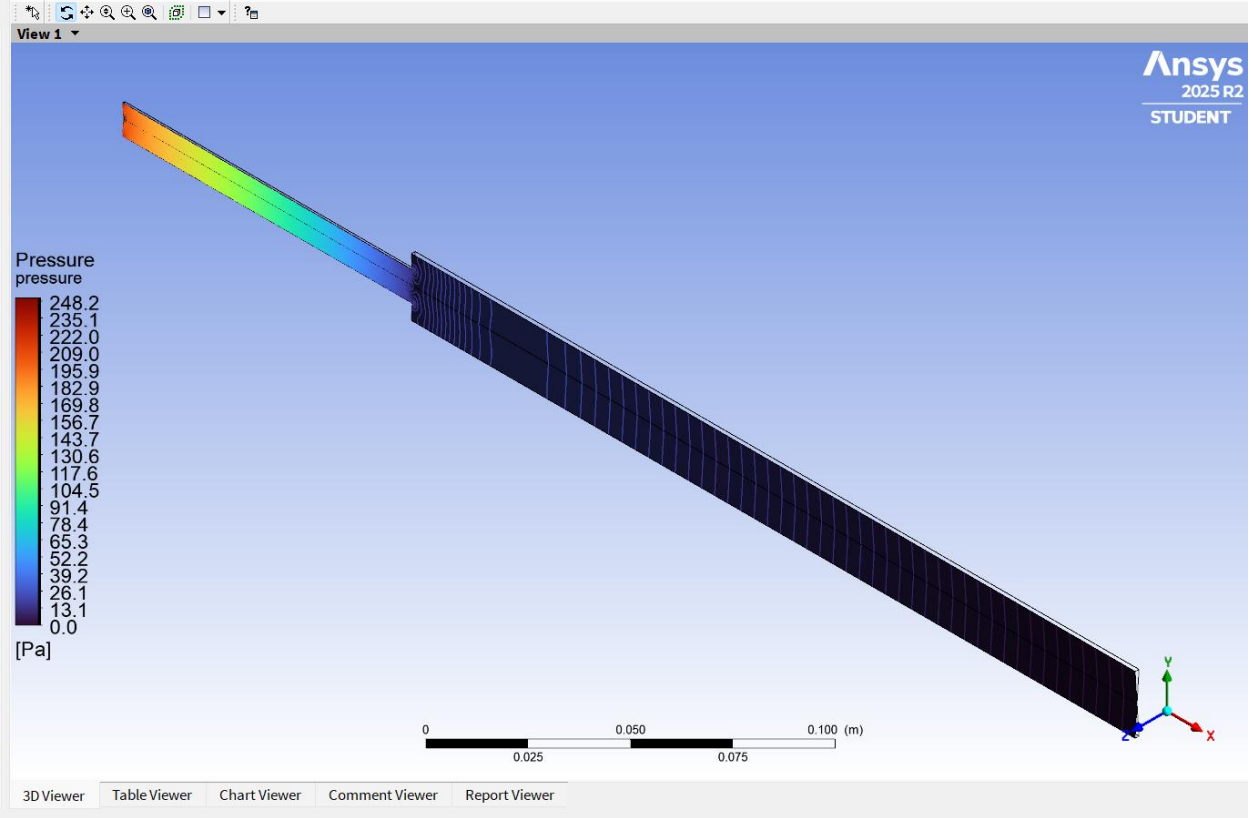


Figure 5. Static pressure contours showing the pressure drop at the expansion step.

A distinct low-pressure region is observed immediately after the sudden expansion ($x = 0$), followed by pressure recovery downstream.

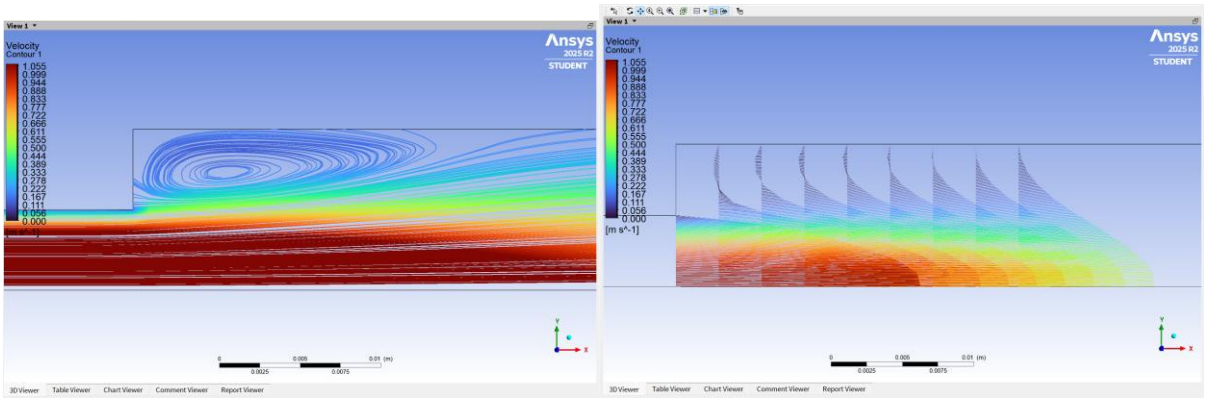
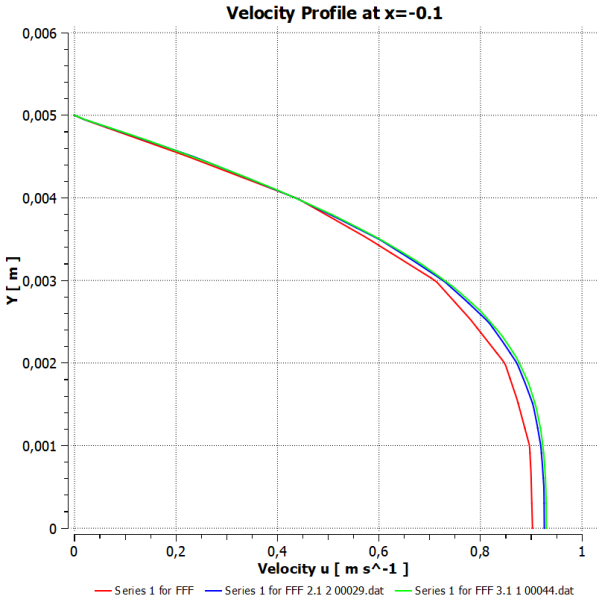


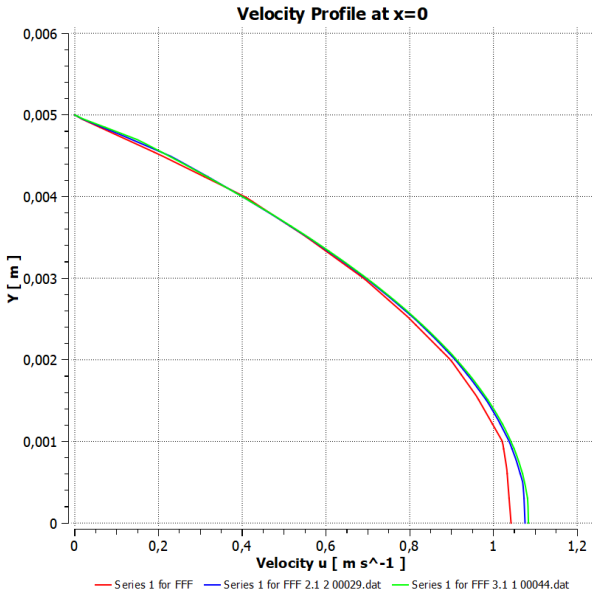
Figure 6 and 7. Streamlines and velocity vectors showing the flow separation and recirculation zone.

Due to the sudden increase in the cross-sectional area which could be seen in the corner of the Figure 7, the fluid is subjected to an adverse pressure gradient. Consequently, the

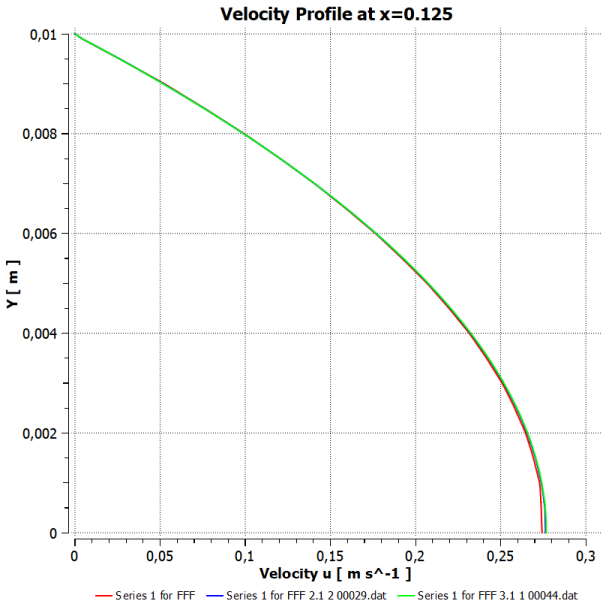
boundary layer separates from the wall immediately at the expansion step (x=0). A free shear layer develops between the high-velocity core flow and the stagnant fluid near the wall. This interaction induces a primary recirculation zone characterized by negative axial velocities i.e. reverse flow. The flow eventually recovers and re-attaches to the wide pipe wall at the re-attachment point ($x = L_r$).



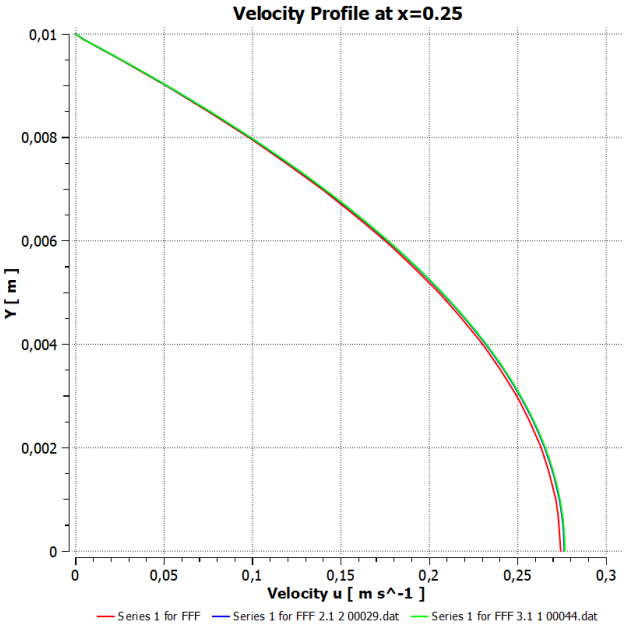
(a)



(b)



(c)



(d)

Figure 8. Axial velocity profiles at various cross-sections: (a) Inlet region, (b) Immediately upstream of the expansion, (c) Immediately downstream of the expansion (recirculation zone), and (d) Outlet cross-section. Comparison between Coarse (Red), Medium (Blue), and Fine (Green) grids.

The grid independence of the numerical solution was evaluated by comparing the axial velocity profiles obtained from three different mesh densities, as shown in Figure 8. In this part how velocities in different cross sections changes.

Inlet and Upstream (a, b): In the fully developed region of the narrow pipe, all three meshes produced identical parabolic velocity profiles, consistent with the analytical Poiseuille flow.

Expansion Region (c): In the critical recirculation zone just after the expansion, where velocity gradients are high, the Medium (Blue) and Fine (Green) grids showed agreement. The Coarse (Red) grid also captured the general flow features but exhibited slight deviations in the shear layer.

Outlet (d): Far downstream, the flow re-developed into a parabolic profile, and all grids converged to the same solution.

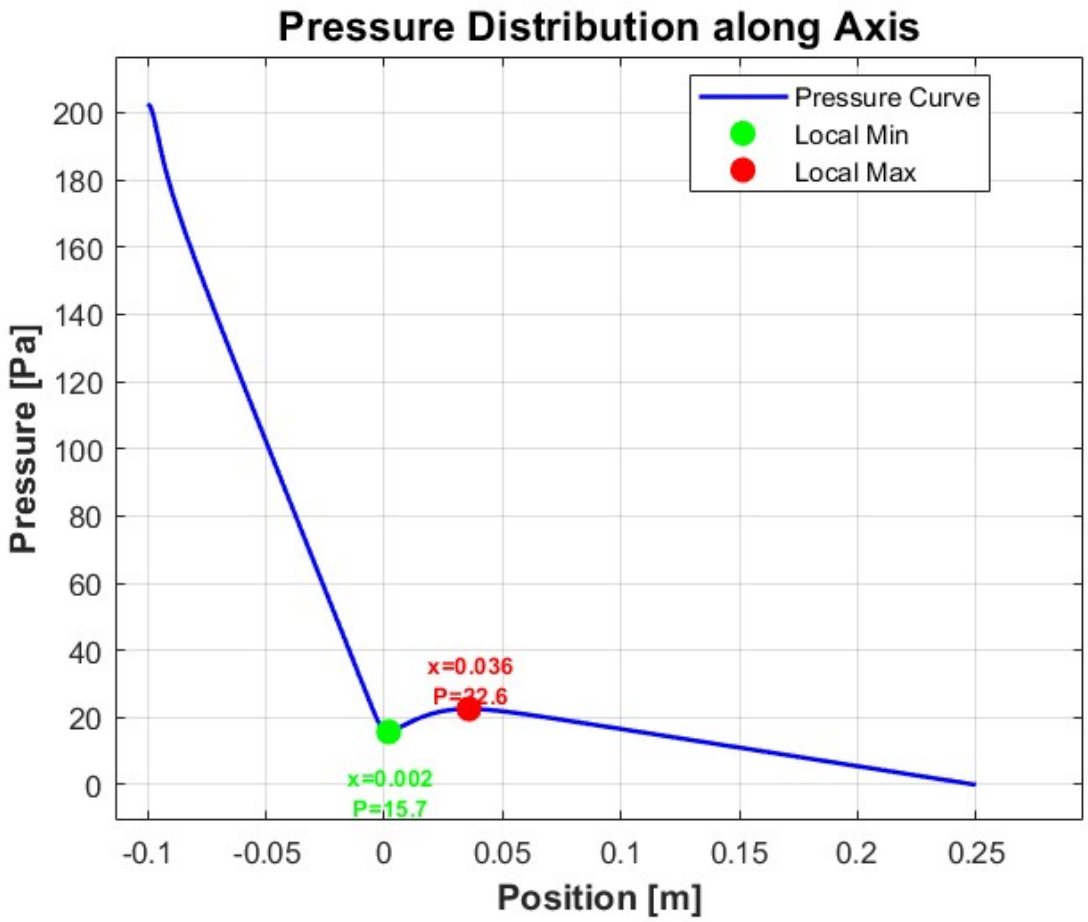


Figure 9: Static pressure distribution along the centerline (y=0) of the pipe. The locations of the local pressure minimum P_{min} immediately after the expansion and the recovered pressure maximum P_{max} are highlighted.

To analyze the pressure recovery along the pipe, static pressure data was extracted along the symmetry axis. The data points were exported from ANSYS CFD-Post in CSV format and post-processed using MATLAB. A custom MATLAB script was developed to import the pressure-position data. Identify the local minimum pressure within the recirculation zone. Identify the local maximum pressure downstream where pressure recovery is achieved. Annotate these critical points on the pressure distribution curve as shown in Figure X.

The pressure distribution along the centerline reveals the flow dynamics caused by the sudden expansion at x=0. As the fluid enters the wide section, the flow separates, creating a low-pressure region. The pressure reaches a local minimum of $P_{min} = 15.7$ Pa at $x = 0.002\ m$,

which corresponds to the core of the recirculation zone. Downstream of the expansion, the flow reattaches to the wall and decelerates, converting kinetic energy back into static pressure (Bernoulli principle). This "pressure recovery" region ends at a local maximum of $P_{max} = 22.6$ Pa at $x \approx 0.036$ m. After the recovery peak, the pressure decreases linearly due to frictional losses in the fully developed pipe flow.

3.1.3 Quantitative Validation

Comparison with Analytical Solution

The numerical velocity profile in the fully developed region of the narrow pipe was compared with the analytical Poiseuille solution given by:

$$u(r) = 2V_{avg}\left(1 - \frac{r^2}{R^2}\right.$$

To validate the prediction of the recirculation zone, a parametric study was conducted by varying the Reynolds number from 25 to 200. The re-attachment length (L_r) was measured for each case, and the dimensionless length was plotted against the Reynolds number

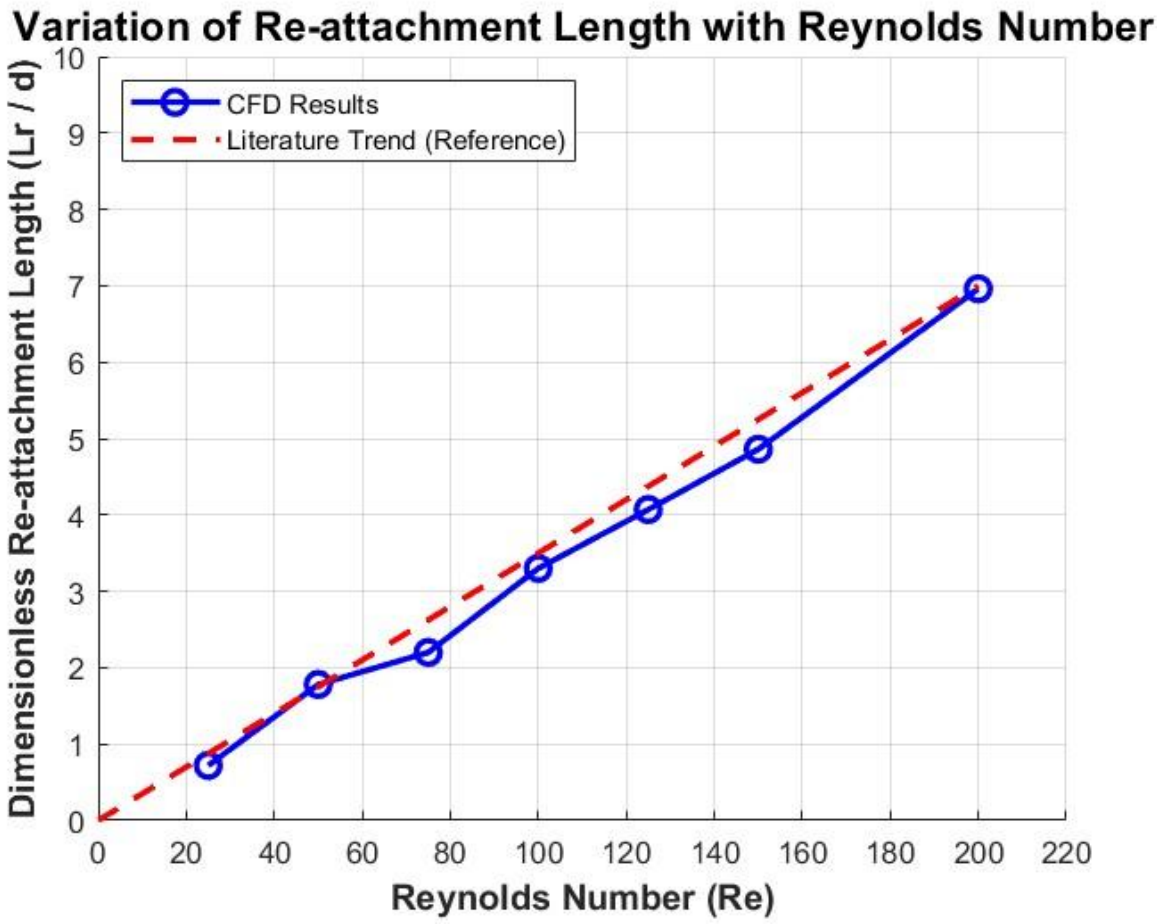


Figure 10: Variation of dimensionless re-attachment length with Reynolds number.

The results demonstrate a linear relationship between the re-attachment length and the Reynolds number. As shown in Figure 6 The numerical velocity profile is compared with the analytical solution for fully developed laminar pipe flow (Poiseuille flow) as described in White [3]. The blue data points are manually taken from shear wall u / X distance charts where wall shear is 0 it was noted for 7 points. These data points are in the Table 1.

Re	25	50	75	100	125	150	200
X(m)	0,0072	0,0178	0,022	0,033	0,0407	0,0486	0,0696

Table 1: Data points for L_r of different Re

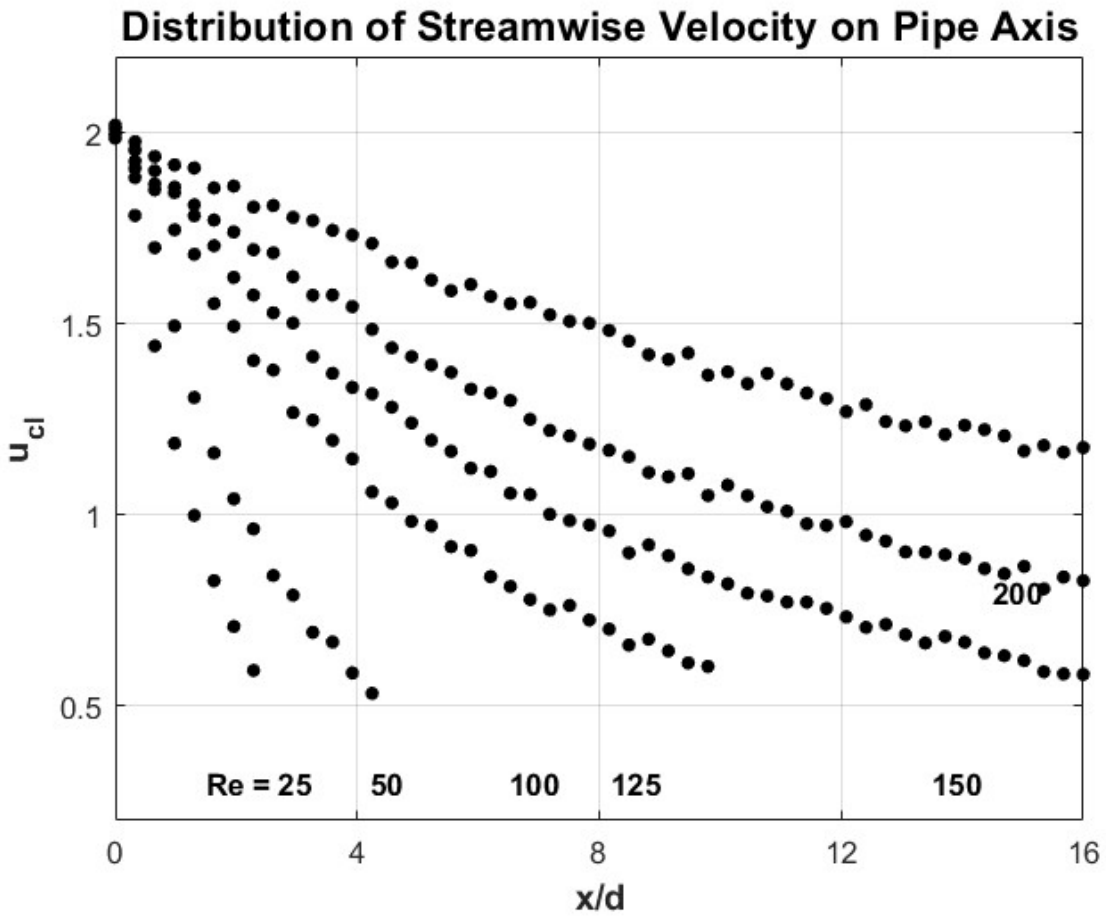


Figure 10: Distribution of Streamwise Velocity on Pipe Axis using different RE values

From the figure X we could match the results with experimental data U profiles for central point.

3.2 Task 2: Turbulent Flow ($Re = 10^5$)

In the second task, the Reynolds number was increased to Re = 10^5 to simulate fully turbulent flow. To achieve this high Reynolds number while maintaining incompressibility (low Mach number), the dynamic viscosity was reduced to $mu = 0.0001\ kg/m \cdot s$, and the inlet velocity was set to $V_{inlet} = 10\ m/s$. Two standard RANS (Reynolds-Averaged Navier-Stokes) turbulence models were employed to solve the flow field:

1. **Standard k-e Model:** Uses standard wall functions.

2. **Standard k-w Model:** Solves for turbulence kinetic energy (k) and specific dissipation rate (ω), providing better resolution in the near-wall region. [2]

3.2.1 Qualitative Comparison (Flow Field)

The pressure and velocity fields obtained from both models were compared qualitatively. Figure 7 illustrates the static pressure distribution in the vicinity of the expansion.

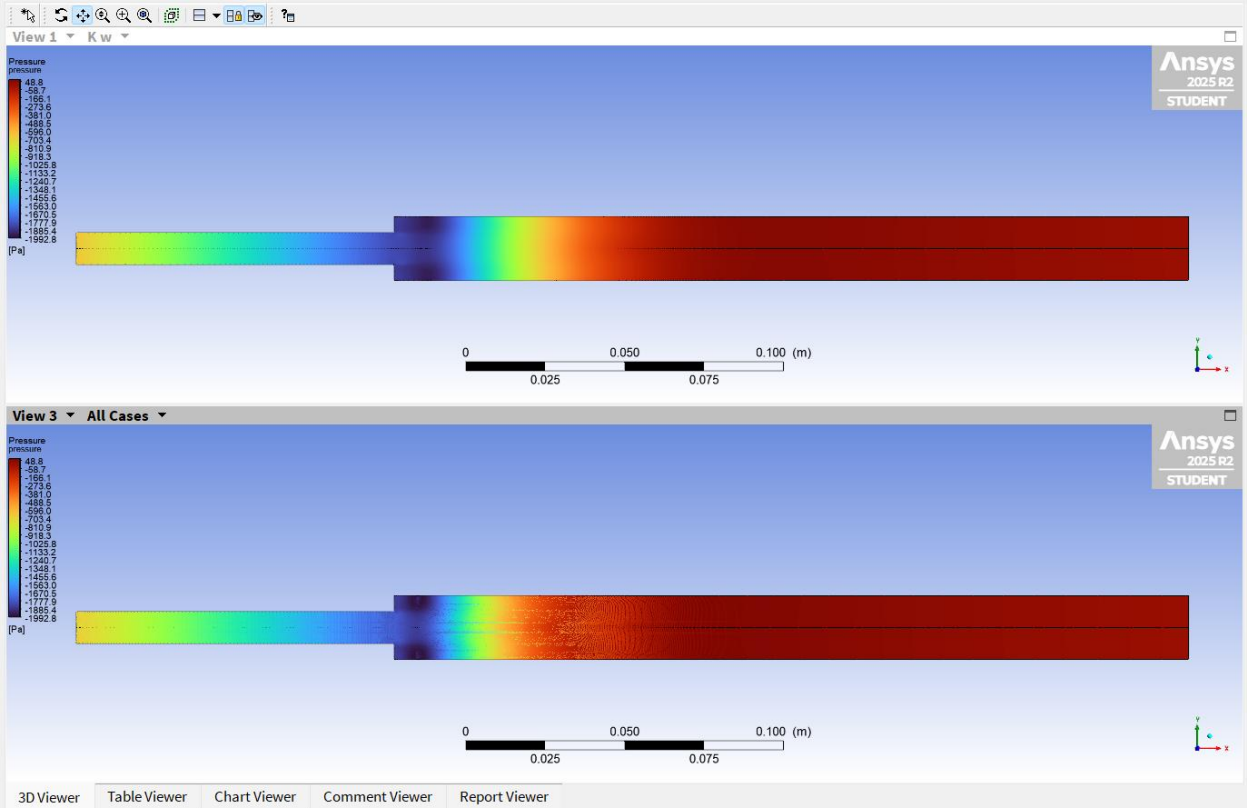


Figure 11: Comparison of static pressure contours for k-epsilon (top) and k-omega (bottom) models.

Both models predict a significant pressure drop immediately after the expansion corner, followed by pressure recovery downstream. However, the low-pressure core in the recirculation zone appears slightly different in size between the two models, suggesting a difference in the predicted vortex length.

The velocity contours illustrate the flow acceleration in the narrow pipe (red region, $V \approx$ 11 m/s) and the subsequent deceleration in the wide pipe. A distinct low-velocity recirculation

zone (dark blue region) is observed immediately downstream of the expansion step. Although the general flow patterns are similar, subtle differences in the length and shape of the low-velocity core can be seen between the two models, consistent with the re-attachment length quantitative data.

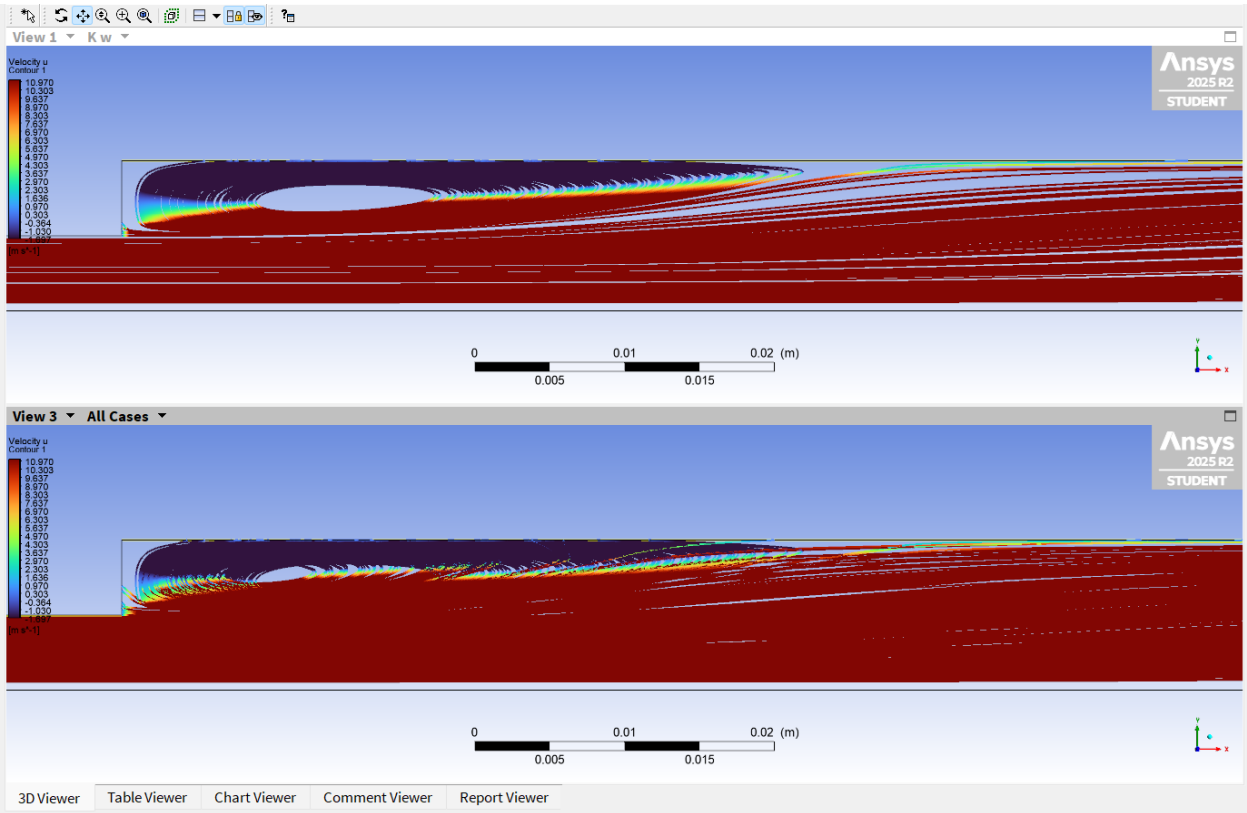


Figure 12: Streamlines showing the recirculation zone for k-epsilon (top) and k-omega (bottom) models.

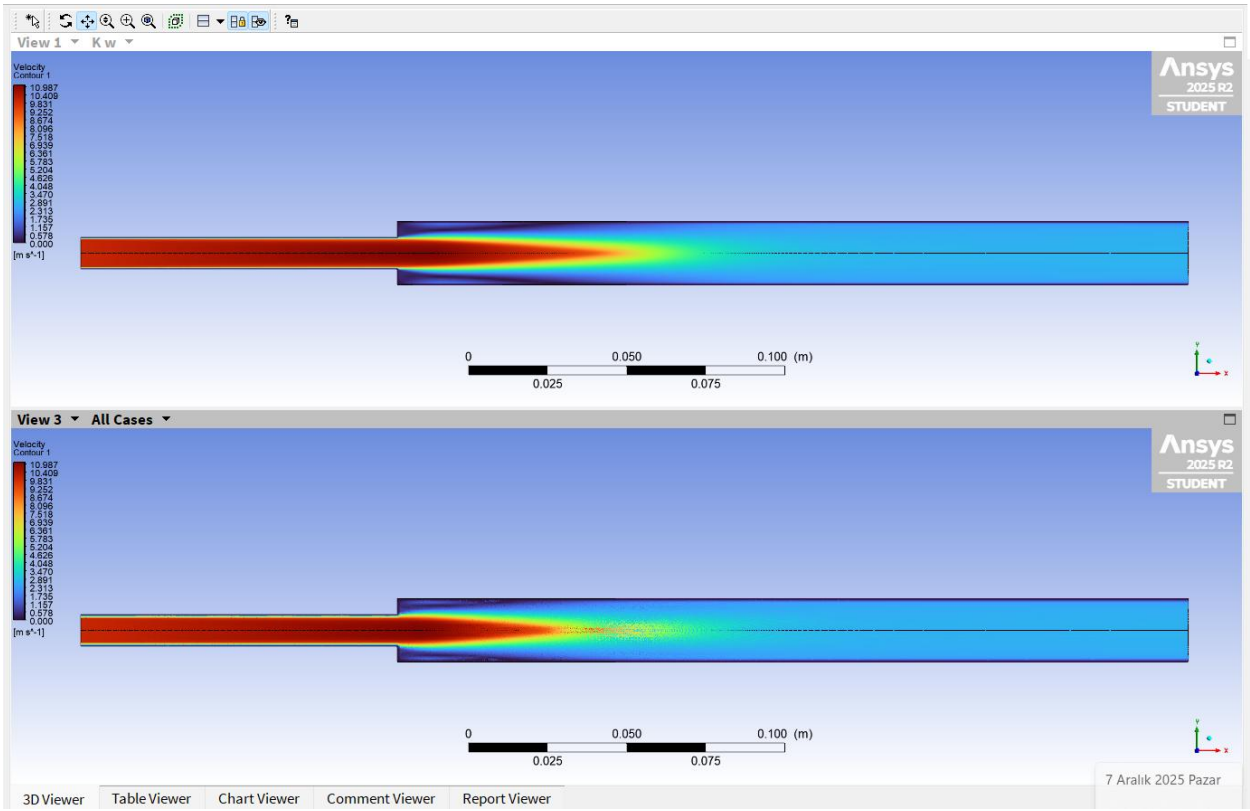


Figure 13: Velocities showing the recirculation zone for k-epsilon (top) and k-omega (bottom) models..

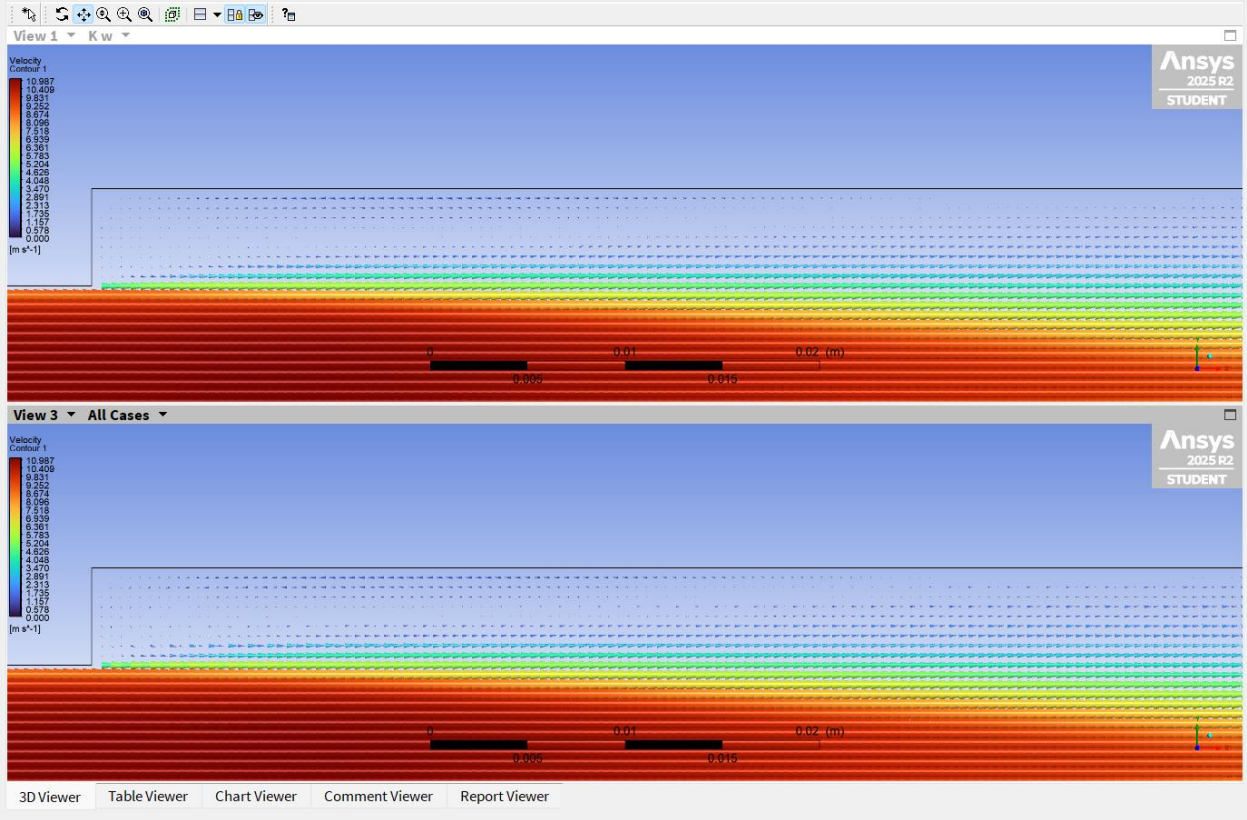


Figure 14: Close-up view of velocity vectors colored by velocity magnitude for turbulent flow $Re = 10^5$. Top: Standard $k - \epsilon$ model, Bottom: Standard $k - \omega$ model.

The numerical simulation of turbulent flow at $Re = 10^5$ revealed distinct differences between the predictions of the Standard $k - \epsilon$ *and Standard* $k - \omega$ models. A comprehensive analysis of velocity contours, pressure fields, streamlines, and wall shear stress distributions leads to the following observations: Both models captured the gross features of the sudden expansion flow, including the flow acceleration in the narrow pipe, separation at the step, and the development of a recirculation zone. As shown in the Velocity Contours (Figure 14), the high-momentum core jet dissipates as it expands into the wide pipe. The Pressure Contours (Figure 11) further confirm this behavior, showing a sharp pressure drop at the expansion plane followed by a gradual pressure recovery region, consistent with the Borda-Carnot pressure loss mechanism. The most critical discrepancy between the models lies in the prediction of the re-attachment length (L_r).

3.2.2 Quantitative Comparison (Re-attachment Length)

To precisely determine the re-attachment length (L_r) and compare the accuracy of the models, the wall shear stress (τ_w) along the bottom wall of the wide pipe was plotted against the axial distance (x).

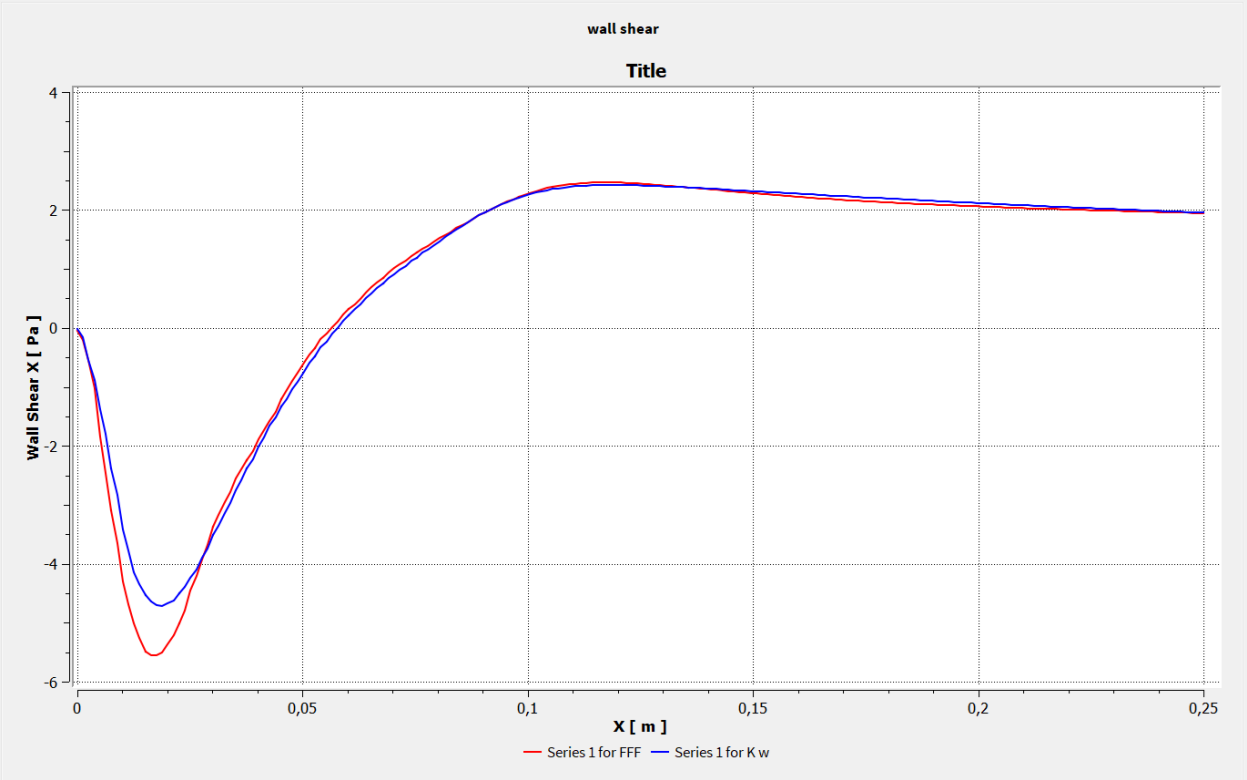


Figure 15: Distribution of axial wall shear stress along the wide pipe wall for k-epsilon and k-omega models.

As seen in Figure 15, the wall shear stress is negative within the recirculation zone meaning reverse flow and becomes positive after the re-attachment point. The point where the curve crosses zero ($\tau_w = 0$) represents the re-attachment point.

The Standard k $-\epsilon$ model predicted a shorter recirculation zone ($L_r \approx 0.052$ m). This model is known to over-predict the turbulent kinetic energy in stagnation and separation regions, which leads to excessive mixing and, consequently, an earlier re-attachment of the flow. The Standard k-w model predicted a significantly longer recirculation zone ($L_r \approx 0.053$ m). This model is formulated to resolve the viscous near-wall region directly without relying on damping

functions. Therefore, it is inherently more accurate in predicting boundary layer separation and re-attachment in flows subjected to adverse pressure gradients.

Table 2 summarizes the re-attachment lengths predicted by both models.

Turbulence Model, Re	attachment Point (x),	Recirculation Length (L_r)
Standard k- epsilon	0.052 m,	0.052 m
Standard k- omega	0.053 m	0.053 m

Table 2: Comparison of re-attachment lengths (L_r) predicted by turbulence models.

(Note: Since the expansion step is located at x=0 in the plot, $Lr = \text{x_reattachment}$).

3.2.3 Discussion on Turbulence Models

This difference can be attributed to the mathematical formulation of the models:

- The k-epsilon model uses "Wall Functions" to bridge the viscosity-affected region between the wall and the fully turbulent region. It is generally robust for free-shear flows but tends to under-predict the size of separation bubbles in flows with adverse pressure gradients.
- The k-omega model does not rely on wall functions and integrates the solution through the viscous sub-layer (provided the mesh is fine enough). Therefore, it is generally considered more accurate for predicting boundary layer separation and re-attachment points.

Based on literature trends for sudden expansion flows, the k-omega model is expected to provide a prediction closer to experimental reality for this specific problem type.

3.3 Task 3: Pressure Loss Analysis

This section investigates the minor head losses associated with flow expansion. The pressure loss coefficient (K) was computed for various expansion ratios (d/D) and geometries (sudden vs. gradual expansion).

3.3.1 Methodology: Parametric Study

To efficiently analyze the effect of geometric variations on pressure loss, a Parametric Study was conducted within the ANSYS Workbench environment. Instead of manually recreating the geometry and setup for each case, the following automated workflow was implemented:

Input Parameters: The wide pipe radius (R_2) in Design Modeler was defined as an Input Parameter. This allowed the expansion ratio (d/D) to be varied systematically by changing the radius R_2 while keeping the inlet radius ($R_1 = 5$ mm) constant.

Output Parameters: In ANSYS Fluent, surface monitors were created to calculate the Area-Weighted Average Static Pressure at the Inlet and Outlet boundaries. These monitors were flagged as Output Parameters (P_{inlet} and P_{outlet}).

Design Points: A "Table of Design Points" was populated in Workbench with the required radii corresponding to diameter ratios of $d/D = 0.6, 0.5(Base), 0.4, and\ 0.2$.

Batch Solution: The "Update All Design Points" feature was used to automatically update the geometry, regenerate the mesh, and solve the flow field for each design point sequentially.

The pressure loss coefficient (K) was then calculated for each case using the pressure drop obtained from the parametric table:

$$K = \frac{\Delta p}{\frac{1}{2}\rho V_{inlet}^2} = \frac{P_{inlet} - P_{outlet}}{\frac{1}{2}\rho V_{inlet}^2}$$

where the outlet gauge pressure was set to 0 Pa (P_{outlet} = 0), making the pressure drop equal to the inlet gauge pressure.

3.3.2 Effect of Expansion Ratio (Sudden Expansion)

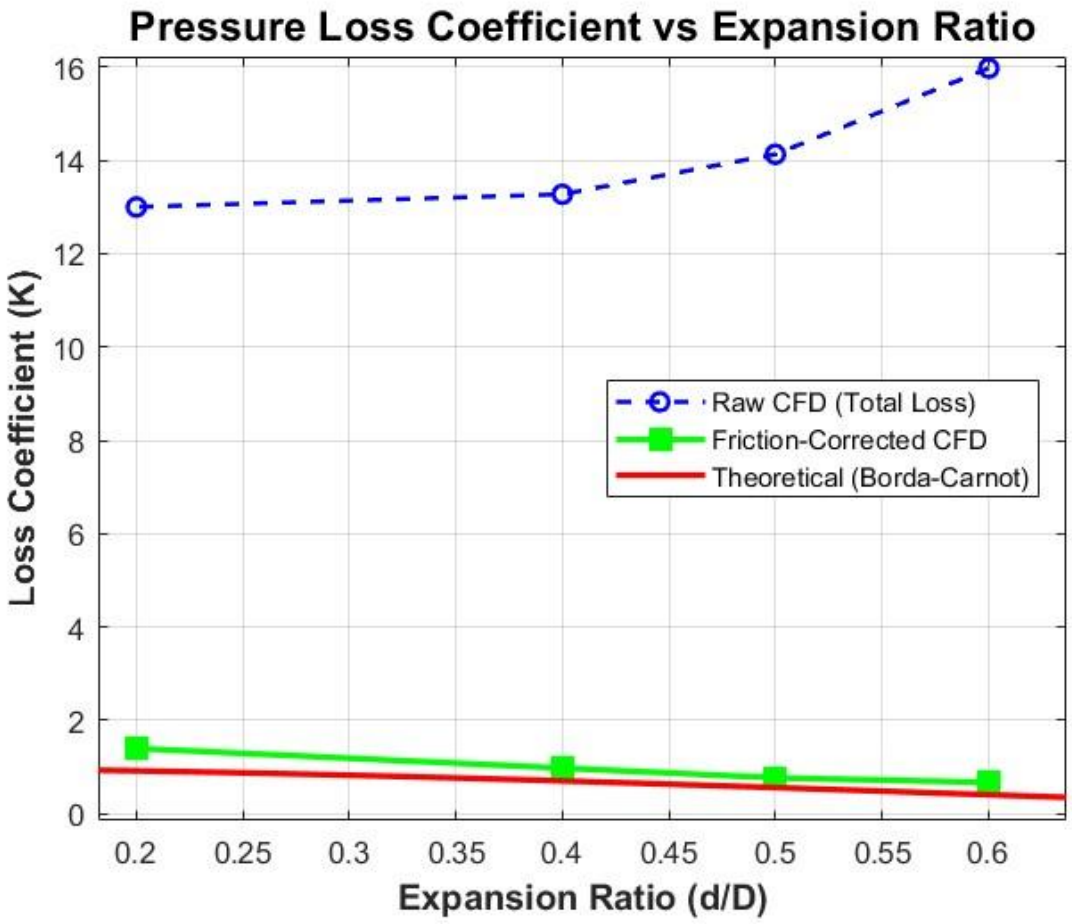


Figure 16. Comparison of pressure loss coefficients (K) versus expansion ratio (d/D). The plot displays the total loss from raw CFD data (blue dashed line), the friction-corrected CFD results (green solid line), and the theoretical Borda-Carnot expansion loss (red solid line).

Table of Design Points

	A	B	C
1	Name	P1 - outlet.radius	P6 - inlet_p-op
2	Units	m	Pa
3	DP 0	0,01	216,83
4	DP 1	0,0083	245,14
5	DP 2 (Current)	0,0125	203,67
6	DP 3	0,025	199,51

Figure 17: For different outlet radiuses Results of P ¡nlet

A difference is observed from Figure 16 between the CFD-calculated loss coefficients ($K_{CFD} \approx 14$) and the theoretical expansion losses ($K_{SE} < 1$). This discrepancy is not an error but a physical consequence of the flow conditions. The theoretical Borda-Carnot formula accounts only for the minor loss at the expansion step. However, the CFD simulation computes the total pressure drop across the entire domain, which includes significant major losses due to friction. Given the low Reynolds number ($Re = 55.4$) and high viscosity, the Darcy friction factor is large ($f = 64/Re \approx 1.15$). Consequently, the frictional loss in the long inlet pipe alone contributes approximately $K \approx f(L/D) \approx 11.5$ to the total coefficient. Therefore, the CFD results correctly reflect that friction is the dominant loss mechanism in this specific setup, dwarfing the minor loss caused by the expansion. Given the laminar flow regime and viscosity used in the simulation, frictional losses are dominant. To isolate the expansion loss component, a friction correction was applied. The theoretical frictional pressure drop was calculated for both pipe sections using the Darcy-Weisbach equation with the laminar friction factor ($f = 64/Re$) and subtracted from the total pressure drop obtained from

$$CFD: K_{corrected} = \frac{\Delta P_{CFD} - \left(\Delta P_{friction,1} + \Delta P_{friction,2}\right)}{\frac{1}{2}\rho V_{inlet}^2}$$

3.3.3 Effect of Gradual Expansion

As illustrated in Figure X, replacing the sudden step with a gradual conical diffuser changes the flow dynamics. Unlike the sudden expansion case, where a massive flow separation and a large recirculation zone were observed (see Figure 4), the gradual expansion allows the fluid to decelerate more smoothly. The moderate divergence angle ($2\theta = 20°$) keeps the adverse pressure gradient within limits that the boundary layer can withstand, thereby maintaining flow attachment along the diffuser walls. The suppression of flow separation eliminates the turbulent mixing losses associated with the eddy formation, resulting in a much lower pressure loss coefficient (K) compared to the sudden expansion geometry.

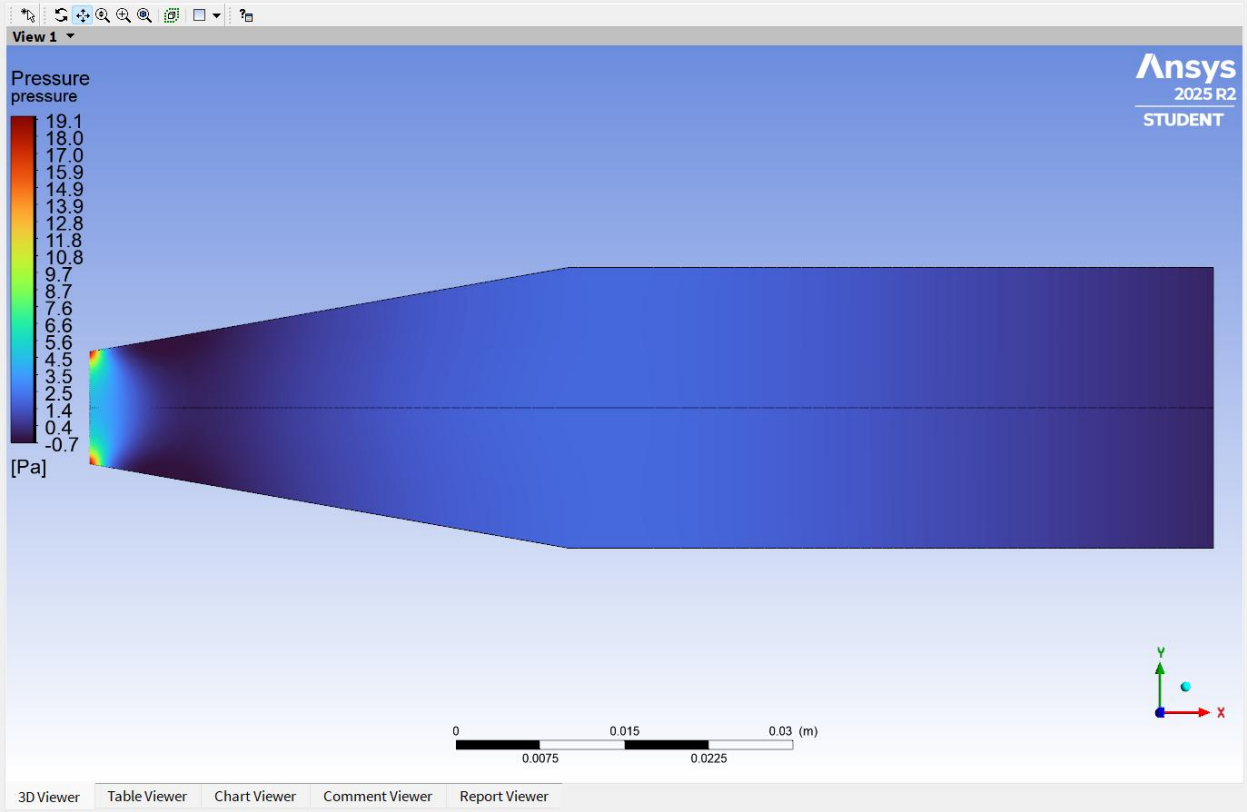


Figure 17: Pressure distribution of gradual expansion body.

To quantify the performance improvement achieved by the gradual expansion, the pressure loss coefficient ($K_{gradual}$) was calculated using the pressure drop (ΔP) obtained from the CFD simulation. The inlet velocity (V_{inlet}) and fluid density (ρ) were kept identical to the sudden expansion case (Re=55.4) to ensure a valid comparison.The loss coefficient is determined by:

$$K_{gradual} = \frac{\Delta P_{diffuser}}{\frac{1}{2}\rho V_{inlet}^2}$$

where: $\Delta P_{diffuser}$: The static pressure difference between the inlet and outlet of the diffuser geometry, extracted from the CFD results using surface monitors .ρ: Fluid density (100 kg/m³).V_{inlet}: Average inlet velocity (0.554 m/s).Using the computed pressure drop of $\Delta P \approx$ 9.995 Pa, the loss coefficient was found to be $K \approx 0.62$. From the table that White provides we could see from the table it should be around 0.35. Also this value is significantly lower than the

sudden expansion case ($K \approx 14$), proving that the conical diffuser geometry effectively minimizes flow separation and irreversible energy losses.

4. Conclusion

In this project, the internal fluid flow through a sudden expansion in a circular pipe was numerically investigated using ANSYS Fluent to analyze the physics of flow separation, validate numerical methods, and evaluate the effects of turbulence modeling and geometric variations on pressure losses. A grid convergence study verified that the numerical solution was mesh-independent, and the laminar flow simulation at $Re = 55.4$ demonstrated excellent agreement with both the analytical Poiseuille solution and experimental centerline velocity data [1]. The parametric study further confirmed that the dimensionless re-attachment length (L_r/d) increases linearly with the Reynolds number, consistent with experimental trends reported in the literature [2]. For the fully turbulent flow case at $Re = 10^5$, the Standard $k - \omega$ model was determined to be more suitable than the Standard $k - \epsilon$ model, as it predicted a longer recirculation zone by more accurately resolving the near-wall region and adverse pressure gradients. Finally, the pressure loss analysis revealed that the loss coefficient (K) increases drastically as the expansion ratio (d/D) decreases; however, replacing the sudden expansion with a gradual conical diffuser ($2\theta = 20°$) reduces these energy losses by suppressing the formation of large-scale flow separation.

5. References

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